Arithmetic and Geometric Series – Worksheet
MCR3U

SOLUTIONS

General formula for an arithmetic series:

$$S_n = \frac{n}{2}[2a + (n-1)d] \quad \text{OR} \quad S_n = \frac{n}{2}(a + t_n)$$

General formula for a geometric series:

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

1) Find the designated sum of the arithmetic series

a) S_{14} of $3 + 7 + 11 + 15 + \cdots$ (+4)

$S_{14} = \frac{14}{2}[2(3) + (14-1)(4)]$

$= 7[6 + 13(4)]$

$= 406$

b) S_{11} of $-13 - 11 - 9 - 7 - \cdots$ (+2)

$S_{11} = \frac{11}{2}[2(-13) + (11-1)(2)]$

$= 5.5[-26 + 10(2)]$

$= -33$

c) S_9 of $22 + 20 + 18 + 16 + \cdots$ (−2)

$S_9 = \frac{9}{2}[2(22) + (9-1)(-2)]$

$= 4.5[44 + 8(-2)]$

$= 126$

d) S_{35} of $-2 - 5 - 8 - 11 - \cdots$ (−3)

$S_{35} = \frac{35}{2}[2(-2) + (35-1)(-3)]$

$= 17.5[-4 + 34(-3)]$

$= -1855$

2) Determine the sum of each arithmetic series

a) $6 + 13 + 20 + \cdots + 69$ (+7)

$t_n = a + (n-1)d$

$69 = 6 + (n-1)(7)$

$63 = (n-1)(7)$

$9 = n - 1$

$n = 10$

$S_{10} = \frac{10}{2}(6 + 69)$

$= 5(75)$

$= 375$

b) $4 + 15 + 26 + \cdots + 213$ (+11)

$213 = 4 + (n-1)(11)$

$209 = (n-1)(11)$

$19 = n - 1$

$n = 20$

$S_{20} = \frac{20}{2}(4 + 213)$

$= 10(217)$

$= 2170$

c) $5 - 8 - 21 - \cdots - 190$ (−13)

$-190 = 5 + (n-1)(-13)$

$-195 = (n-1)(-13)$

$15 = n - 1$

$n = 16$

$S_{16} = \frac{16}{2}(5 - 190)$

$= 8(-185)$

$= -1480$

d) $100 + 90 + 80 + \cdots - 100$ (−10)

$-100 = 100 + (n-1)(-10)$

$-200 = (n-1)(-10)$

$20 = n - 1$

$n = 21$

$S_{21} = \frac{21}{2}(100 - 100)$

$= 0$

3) Find the designated sum of the geometric series

a) S_7 of $4 + 8 + 16 + 32 + \cdots$ (×2)

$S_7 = \frac{4(2^7-1)}{2-1}$

$= 508$

b) S_{13} of $1 - 6 + 36 - 216 + \cdots$ (×−6)

$S_{13} = \frac{1[(-6)^{13}-1]}{-6-1} = 1\,865\,813\,43|$

c) S_{17} of $486 + 162 + 54 + 18 + \cdots$ (×$\frac{1}{3}$)

$S_{17} = \frac{486[(\frac{1}{3})^{17}-1]}{(\frac{1}{3})-1} = 729$

d) S_6 of $3 + 15 + 75 + 375 + \cdots$ (×5)

$S_6 = \frac{3[(5)^6-1]}{5-1} = 11718$

4) Determine S_n for each geometric series

a) $a = 6, r = 2, n = 9$

$S_9 = \frac{6[(2)^9-1]}{2-1} = 3066$

b) $f(1) = 2, r = -2, n = 12$

$S_{12} = \frac{2[(-2)^{12}-1]}{-2-1} = -2730$

c) $f(1) = 729, r = -3, n = 15$

$S_{15} = \frac{729[(-3)^{15}-1]}{-3-1} = 2\,615\,088\,443$

d) $f(1) = 2700, r = 10, n = 8$

$S_8 = \frac{2700[(10)^8-1]}{10-1} = 2.999\,999\,97 \times 10^{10}$

5) If the first term of an arithmetic series is 2, the last term is 20, and the increase constant is +2 …

a) Determine the number of terms in the series

$t_n = a + (n-1)d$

$20 = 2 + (n-1)(2)$

$18 = (n-1)(2)$

$9 = n-1$

$n = 10$

b) Determine the sum of all the terms in the series

$S_{10} = \frac{10}{2}[2 + 20]$

$= 5(22)$

$= 110$

6) A geometric series has a sum of 1365 (S_6). Each term increases by a factor of 4 (r). If there are 6 (n) terms, find the value of the first term.

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$1365 = \frac{a[(4)^6 - 1]}{4 - 1}$$

$$1365(3) = a(4^6 - 1)$$

$$4095 = a(4095)$$

$$a = 1$$

Answers

1) a) 406 **b)** -33 **c)** 126 **d)** -1855

2) a) 375 **b)** 2170 **c)** -1480 **d)** 0

3) a) 508 **b)** 1 865 813 431 **c)** 729 **d)** 11 718

4) a) 3066 **b)** -2730 **c)** 2 615 088 483 **d)** $2.999\,999\,97 \times 10^{10}$

5) a) $n = 10$ **b)** $S_{10} = 110$

6) $t_1 = 1$